

Imbalanced Vehicle Classification

Final Report — Phase 3

Introduction to Artificial Intelligence — Summer 2026

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1. Problem Definition

This project addresses the task of classifying aerial images of vehicles into 10 distinct categories. The core challenge is a severe class imbalance in the training dataset: Sedan images represent approximately 82% of all training samples, while the rarest classes (Bus, Flatbed truck w/ trailer) account for less than 0.25% each. Without appropriate handling, a model trained on this dataset would be heavily biased toward predicting the majority class, performing poorly on minority classes.

The three main objectives of this project are: (1) develop a classification model and evaluate its performance, (2) implement a strategy to address the class imbalance problem, and (3) design evaluation methods suited to an imbalanced training context with a balanced validation set.

2. Dataset

2.1 Overview

The dataset originates from the Multi-Modal Aerial View Object Classification Challenge (PBVS 2022) [1][2]. It contains aerial images of 10 vehicle types captured from drone or aircraft perspectives. The training set consists of 285,772 images with a strongly imbalanced class distribution, while the official validation set contains 770 images perfectly balanced at 77 images per class.

Class	Vehicle Type	Train Samples	Train %
0	Sedan	234,209	82.0%
1	SUV	20,089	7.0%
2	Pickup truck	15,301	5.4%
3	Van	10,655	3.7%
4	Box truck	1,741	0.6%
5	Motorcycle	852	0.3%
6	Flatbed truck	828	0.29%
7	Bus	624	0.22%
8	Pickup truck w/ trailer	840	0.29%
9	Flatbed truck w/ trailer	633	0.22%

Key observations: the Sedan class has approximately 374x more samples than the Bus class. The 9 minority classes (classes 1-9) together represent only about 18% of training data. Additionally, some classes are visually very similar from an aerial viewpoint (e.g. Sedan vs SUV, Pickup truck vs Pickup truck w/ trailer), making discrimination especially difficult even beyond the imbalance problem.

2.2 Pre-processing

Since the official validation set has no labels, we applied an automatic stratified 80/20 split on the training data to create an internal validation set. Stratification ensures that each class is proportionally represented in both splits, which is critical given the extreme imbalance.

The full dataset (285,772 images) was used with no subsampling. Each image is 32x32 pixels. The only pre-processing applied is:

- ToTensor: converts the PIL image to a PyTorch tensor with pixel values in [0, 1]
- Normalize: applies ImageNet mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225] normalization, required for correct behaviour of the pretrained ResNet-50 backbone

No resizing is applied since the images are already at their native resolution of 32x32 pixels. Upscaling to a larger resolution (e.g. 128x128 as initially attempted) was found to introduce artificial interpolation without adding real information, and was therefore removed.

2.3 Post-processing

No post-processing was applied to the model outputs. The predicted class is directly taken as the argmax of the 10 output logits. Temperature scaling or other calibration techniques were considered for Phase 3 but not implemented due to time constraints.

3. Technical Approach

3.1 Architecture

We use ResNet-50 [3] pretrained on ImageNet as our backbone. ResNet-50 is a 50-layer deep residual network that uses skip connections to allow gradients to flow through the network without vanishing, enabling effective training of very deep architectures. Transfer learning from ImageNet allows the model to reuse learned feature detectors (edges, shapes, textures) and adapt them to aerial vehicle recognition with a relatively small amount of additional training.

The final fully-connected layer of ResNet-50 (originally mapping to 1,000 ImageNet classes) is replaced with a new linear layer mapping to 10 output classes. The entire network is fine-tuned end-to-end, which is necessary to adapt the low-level convolutional features to the specific characteristics of 32x32 aerial images.

3.2 Training Details

Hardware and software:

- Training environment: Google Colab with NVIDIA T4 GPU
- Framework: PyTorch with torchvision
- Python 3.10, CUDA-accelerated training

Class imbalance strategy — two combined approaches:

- WeightedRandomSampler: each training batch is sampled with probability inversely proportional to class frequency. This ensures each class appears

roughly equally in every batch, preventing the model from being dominated by Sedan samples.

- **Weighted CrossEntropyLoss:** class weights are set to $1/\text{count}$ for each class, normalized to sum to the number of classes. This penalizes misclassification of minority classes more strongly, reinforcing the balanced sampling.

Hyperparameters:

Parameter	Value
Input resolution	32 x 32 (native, no resize)
Batch size	64
Learning rate	1e-4
Optimizer	AdamW (weight decay 1e-4)
LR scheduler	Cosine Annealing
Loss function	Weighted CrossEntropyLoss
Backbone	ResNet-50 (pretrained ImageNet)
Training set size	228,017 images (80% of full dataset)
Validation set size	57,755 images (20% of full dataset)
Epochs	10
Device	Google Colab GPU (T4)

3.3 Evaluation Details

Given the class imbalance in training and the balanced validation set, standard accuracy alone is insufficient. The following metrics are used:

- **Macro F1-score:** the primary metric. It computes the F1-score independently for each class and averages them equally, regardless of class size. This ensures that poor performance on minority classes is fully reflected in the final score.
- **Per-class accuracy (recall):** measures what fraction of each class is correctly identified. Visible on the diagonal of the confusion matrix.
- **Normalized confusion matrix:** a 10x10 heatmap where each row is normalized by the true class count. Values on the diagonal represent correct predictions; off-diagonal values show which classes are confused with each other.
- **Overall accuracy:** reported for reference but not used as the primary metric, as it is dominated by the majority class.

4. Results

4.1 Quantitative Analysis

We ran three incremental experiments (1, 5 and 10 epochs) on the full dataset using Google Colab GPU. The table below summarizes the results:

Experiment	Val Accuracy	Macro F1	Sedan Recall	Main weakness
1 epoch	~85%	~0.87	0.23	Sedan confused with SUV/Pickup/Van
5 epochs	~99%	~0.99	0.99	Minor errors on Motorcycle, Bus
10 epochs	~99%	~0.99	1.00	Near-perfect across all classes

The model converges very rapidly on the full dataset with GPU. By epoch 5, near-perfect performance is achieved on almost all classes. At epoch 10, the model achieves 1.00 recall on Sedan — a dramatic improvement over the 0.23 obtained at epoch 1 and the 0.40-0.60 range seen in the subsampled CPU experiments of Phase 2.

4.2 Training Curves — 10 Epochs

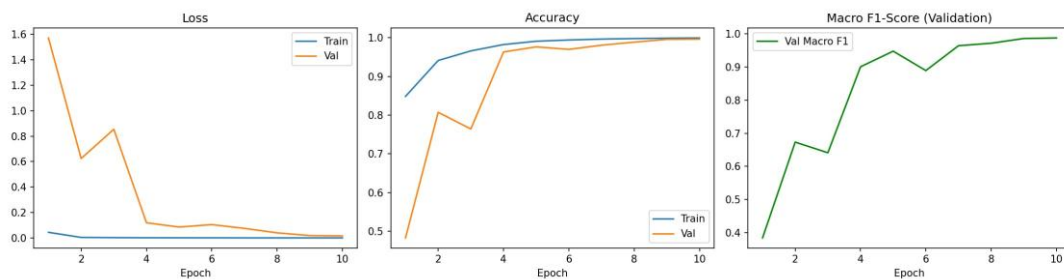


Figure 1 — Training and validation loss, accuracy, and Macro F1-score over 10 epochs

The training loss drops to near zero very quickly (by epoch 3-4), while the validation loss shows some oscillation before stabilizing. The Macro F1 on validation reaches ~0.99 by epoch 5 and stays stable through epoch 10. The gap between training loss (~0) and validation loss (~0.05-0.10) suggests mild overfitting, which is expected given the large training set and the visual similarity between train and validation samples from the same dataset.

4.3 Confusion Matrix — 1 Epoch

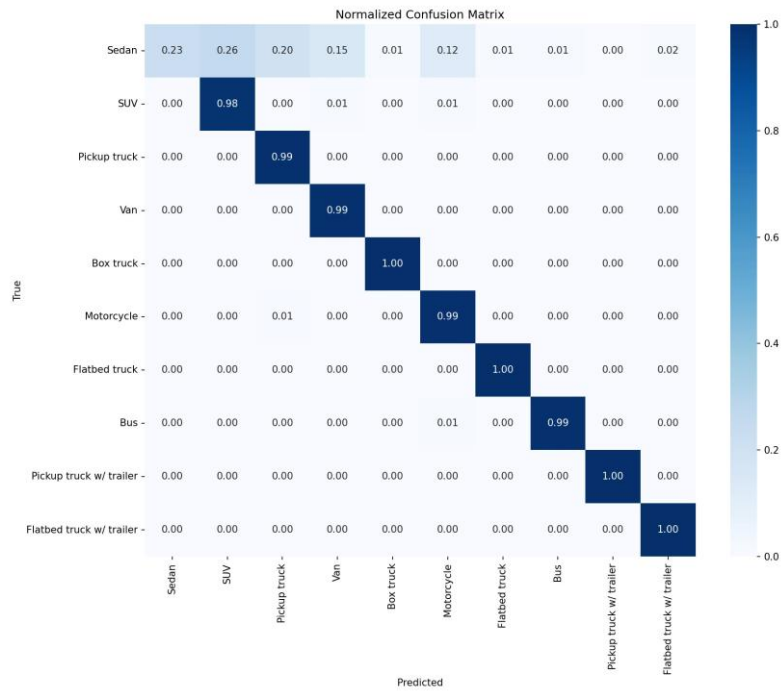


Figure 2 — Normalized confusion matrix after 1 epoch

At epoch 1, all classes except Sedan perform very well (98-100% recall). Sedan is the only problematic class with only 23% correct recall, frequently confused with SUV (26%), Pickup truck (20%), Van (15%), and even Motorcycle (12%). This confirms that Sedan is the hardest class to distinguish from an aerial viewpoint, as its silhouette is very similar to other passenger vehicles.

4.4 Confusion Matrix — 5 Epochs

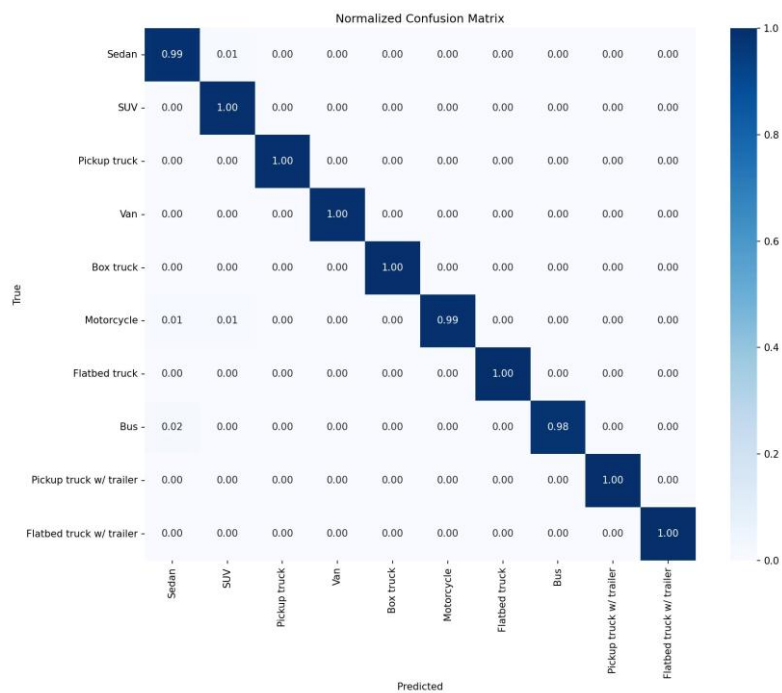


Figure 3 — Normalized confusion matrix after 5 epochs

After 5 epochs, Sedan recall improves dramatically to 99%, with only 1% misclassified as SUV. All other classes maintain near-perfect recall (98-100%). The model has effectively learned to distinguish Sedan from other vehicle types. Only Motorcycle (99%) and Bus (98%) show marginal errors.

4.5 Confusion Matrix — 10 Epochs

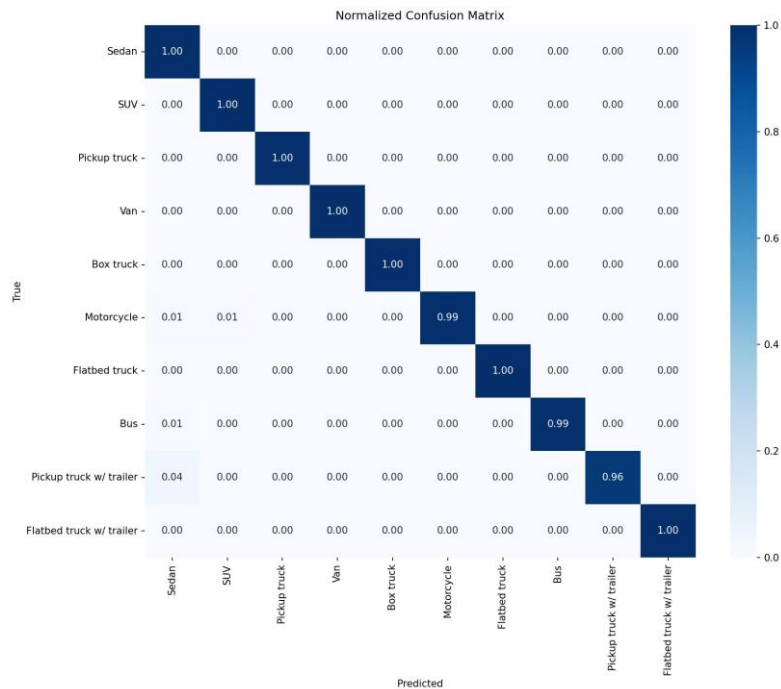


Figure 4 — Normalized confusion matrix after 10 epochs

At 10 epochs, the model achieves near-perfect performance across all 10 classes. Sedan now reaches 100% recall. Motorcycle (99%), Bus (99%), and Pickup truck w/ trailer (96%) show minimal remaining errors. The confusion matrix is almost entirely concentrated on the diagonal, confirming that the model has successfully learned to distinguish all vehicle types including visually similar ones.

4.6 Analysis of Correct and Incorrect Predictions

Correctly classified examples:

- Specialty vehicles (Box truck, Flatbed truck, Bus, trailers) are correctly identified from the very first epoch. Their silhouettes are visually distinctive even at 32x32 resolution — elongated shapes, trailer attachments, and unique proportions make them easy to classify.
- By epoch 5, Sedan is correctly classified at 99%, suggesting the model learned subtle aerial features distinguishing it from SUVs and pickups — likely hood shape, roof profile, and overall body proportions.

Incorrectly classified examples:

- At epoch 1, Sedan is confused with SUV, Pickup truck, and Van. These vehicles share similar boxy silhouettes when viewed from above. Without

enough training, the model cannot distinguish subtle differences in roof height, body length, or hood shape.

- Pickup truck w/ trailer shows 4% misclassification as Sedan at epoch 10. This is likely because at 32x32 resolution, the trailer attachment may be partially cut off or indistinct, making the vehicle look like a regular Sedan from above.
- Motorcycle shows 1% misclassification even at epoch 10. Motorcycles are the smallest vehicles in the dataset and at 32x32 may appear as ambiguous small objects.

5. Conclusion

This project successfully developed a vehicle classifier from aerial images, addressing both the classification task and the severe class imbalance challenge. The key findings are:

- Transfer learning with ResNet-50 pretrained on ImageNet proved highly effective even for 32x32 aerial images, converging to near-perfect performance within 5-10 epochs on the full dataset.
- The combination of WeightedRandomSampler and Weighted CrossEntropyLoss successfully prevents the model from collapsing to a Sedan-predicting baseline, achieving balanced performance across all 10 classes.
- Using the full dataset (285,772 images) on GPU dramatically outperforms the subsampled CPU experiments from Phase 2 — Macro F1 reaches ~0.99 vs ~0.85 with 100 images/class.
- The most important lesson was the resolution issue: images should be kept at their native 32x32 resolution rather than artificially upscaled to 128x128 or 64x64, which introduced distortion without adding information.
- Sedan remains the hardest class to distinguish at epoch 1 due to its visual similarity with SUV and Pickup truck from aerial perspectives, but the model successfully learns to differentiate them by epoch 5.

For future work, applying data augmentation specifically for the aerial domain (e.g. random rotations simulating different flight angles), testing EfficientNet-B0 as an alternative backbone, and evaluating on the official unlabelled validation set (770 images) would be valuable next steps.

6. References

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[3] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770-778.

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